Speech Processing

1. Consider a lossless tube model of the vocal tract using two sections of tube. Figure 1.1 shows signal flow graphs of alternative representations of this lossless tube model labelled Structure A and Structure B.

   a) Derive an expression for $\begin{bmatrix} V_1^+ \\ V_1^- \end{bmatrix}$ in terms of $\begin{bmatrix} V_2^+ \\ V_2^- \end{bmatrix}$.

   Then derive a further expression for $\begin{bmatrix} W_1^+ \\ W_1^- \end{bmatrix}$ in terms of $\begin{bmatrix} W_2^+ \\ W_2^- \end{bmatrix}$

   [6]

Solution:
For Structure A:

$$V_2^+ = (1+r_1)V_1^+ + r_1 V_2^-$$
$$\Rightarrow V_1^+ = \frac{1}{1+r_1}\left(V_2^+ - r_1 V_2^-\right)$$
$$V_1^- = z^{-1}\left(-r_1 v_1^+ + (1-r_1)V_2^-\right) = \frac{z^{-1}}{1+r_1}\left(-r_1 V_2^+ + V_2^-\right)$$
$$\begin{bmatrix} V_1^+ \\ V_1^- \end{bmatrix} = \frac{1}{1+r_1}\begin{bmatrix} 1 & -r_1 \\ -r_1 z^{-1} & z^{-1} \end{bmatrix}\begin{bmatrix} V_2^+ \\ V_2^- \end{bmatrix}.$$

For Structure B:

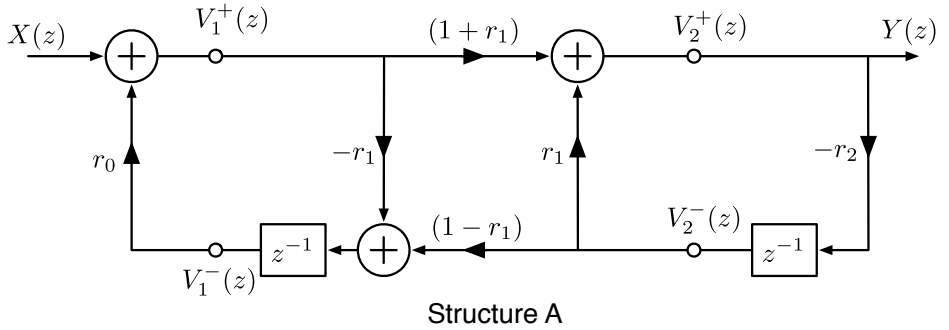
$$W_2^+ = W_1^+ + r_1 W_2^- \Rightarrow W_1^+ = W_2^+ - r_1 W_2^-$$
$$W_1^- = z^{-1}\left(-r_1 W_2^+ + W_2^-\right)$$
$$\begin{bmatrix} W_1^+ \\ W_1^- \end{bmatrix} = \begin{bmatrix} 1 & -r_1 \\ -r_1 z^{-1} & z^{-1} \end{bmatrix}\begin{bmatrix} W_2^+ \\ W_2^- \end{bmatrix}$$

   b) Derive the transfer functions

$$\frac{Y(z)}{X(z)} \text{ and } \frac{Q(z)}{P(z)}$$

   and hence give the relationship between these two transfer functions.

   [6]



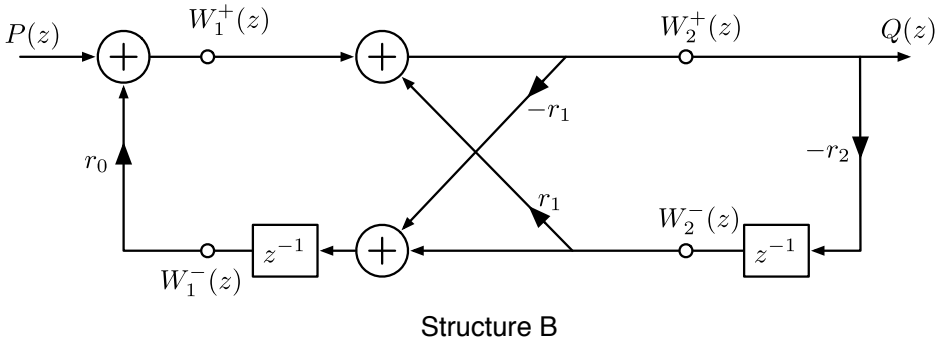


Figure 1.1: Structures representing a lossless tube model

Solution:

Structure A

$$X(1+r_1) = \left(1 + r_1(r_0+r_2)z^{-1} + r_0 r_2 z^{-2}\right) Y.$$

Structure B

$$\begin{aligned} P &= \begin{bmatrix} 1 & -r_0 \end{bmatrix} \begin{bmatrix} 1 & -r_1 \\ -r_1 z^{-1} & z^{-1} \end{bmatrix} \begin{bmatrix} 1 \\ -r_2 z^{-1} \end{bmatrix} Q \\ &= \left(1 + r_1(r_0+r_2)z^{-1} + r_0 r_2 z^{-2}\right) Q. \end{aligned}$$

Hence

$$\frac{Y}{X} = (1+r_1)\,\frac{Q}{P}$$

c) Consider the computations involved in computing one output sample in these two structures. State the number of addition operations needed and the number of multiply operations needed to compute one output sample using Structure A. Repeat this question for Structure B. [4]

Solution:
Structure A requires 6 multiplies and 3 additions.
Structure B requires 4 multiplies and 3 additions.

d) Now consider the two-tube representation given in Structure B. Also consider a sequence of input samples $\{p(n)\}$ and a corresponding sequence of output samples $\{q(n)\}$.

Write out the software programming code that computes the output sequence $\{q(n)\}$ given an input sequence $\{p(n)\}$. You may use the syntax of MATLAB or any other language or pseudo-code providing it is clearly set out.

Hint: Use one FOR loop in your code to count through each sample.

Note that marks will not be deducted for syntax errors in the code providing that the operations are described unambiguously. It is not necessary to show any variable declarations or intialization in the code providing they are clear from the context. [4]

Solution:

```
for n=1:N
pp = p(n);
wp = pp + r(0)*wm(1);
wp = wp + r(1)*wm(2);
wm(1) = wm(2) - r(1)*wp;
q(n) = wp;
wm(2) = -r(2)*wp;
end
```

2. Linear prediction is applied to a speech signal $s(n)$. The order of the LPC is chosen to be p. The resulting prediction coefficients are denoted a_k for $k = 1, \ldots, p$ and the LPC system function is given by

$$H(z) = \frac{1}{1 - \sum_{k=1}^{p} a_k z^{-k}}.$$

a) Why is LPC generally considered to be suitable for encoding speech signals? Include in your answer any relevant properties of speech signals. [4]

Solution:
Speech formants arise due to acoustic resonances in the vocal tract and these can be modelled using a complex conjugate pole pair per resonance. Resonances, and their associated harmonic structure, are predictable with high accuracy using an auto-regressive structure - the key structure of LPC. LPC can be computationally efficient.

b) Consider (i) voiced and (ii) unvoiced speech. Briefly describe the characteristics of the prediction residual in the case when the prediction is exact. [2]

Solution:
If the prediction of $H(z)$ is exact, then the prediction residual will be an impulse train for voiced speech and white noise for unvoiced speech.

c) Summarise the method of prediction of pitch periodicity in LPC-based speech coding methods. [2]

Solution:
Notably, LPC does not predict the quasi-periodic nature of pitch periodicity. This is handled separately in speech coders by a long-term predictor (sometimes called the adaptive codebook).

d) Let i, k and m be indices. Consider a particular analysis frame containing a segment of the speech signal $s(n)$, and denote the corresponding prediction error $e(n)$. Now define

$$\varphi(i,k) = \sum_{m} s(m-i)s(m-k).$$

i) For this analysis frame, give the expression for the squared prediction error ε in terms of a_k and s. [2]

ii) Show how the prediction coefficients a_k can be chosen to minimize ε and write an expression for ε in terms of φ and a_k. [4]

Solution:
(i)

$$\varepsilon = \sum_m e^2(m) = \sum_m \left(s(n) - \sum_{k=1}^{p} a_k s(m-k) \right)^2$$

(ii)
From the results of setting the partial derivate w.r.t. to a_k to 0, the minimum ε is obtained when a_k are chosen as solutions to

$$\sum_m s(m-i)s(m) = \sum_{k=1}^{p} a_k \sum_m s(m-i)s(m-k)$$
$$1 \leq i \leq p.$$

Next we can write for optimal $a_k = \tilde{a}_k$

$$\begin{aligned}\varepsilon &= \sum_m \left(s(n) - \sum_{k=1}^{p} a_k s(m-k) \right)^2 \\ &= \sum_m s^2(m) - \sum_{k=1}^{p} \tilde{a}_k \sum_m s(m)s(m-k)\end{aligned}$$

and using

$$\varphi(i,k) = \sum_m s(m-i)s(m-k)$$

leads to

$$\varepsilon = \varphi(0,0) - \sum_{k=1}^{p} \tilde{a}_k \varphi(0,k).$$

e) Now consider a particular frame of speech for which the first 3 values of the autocorrelation function are 51, 45, 39. Determine the optimal linear prediction coefficients in $H(z)$ for 2nd order autocorrelation LPC and find the corresponding minimum squared error. [6]

Solution:
In this case we can use the autocorrelation method for LPC formulated as

$$\begin{bmatrix} R(0) & R(1) \\ R(1) & R(0) \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} R(1) \\ R(2) \end{bmatrix}.$$

The autocorrelation matrix is $\begin{bmatrix} 51 & 45 \\ 45 & 51 \end{bmatrix}$ which has determinant 576 and inverse $\begin{bmatrix} 0.089 & -0.078 \\ -0.078 & 0.089 \end{bmatrix}$.

Hence $\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 0.089 & -0.078 \\ -0.078 & 0.089 \end{bmatrix} \begin{bmatrix} 45 \\ 39 \end{bmatrix} = \begin{bmatrix} 0.963 \\ -0.039 \end{bmatrix}$. Thus we can write

$$H(z) = \frac{1}{1 - 0.963z^{-1} + 0.039z^{-2}}.$$

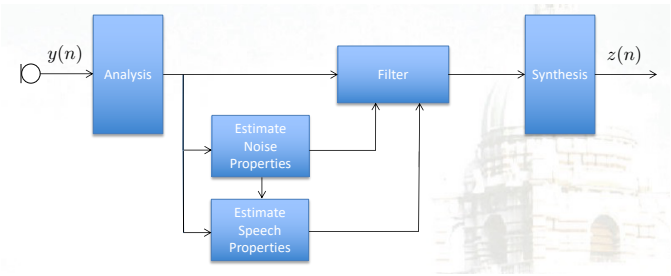
The minimum mean squared prediction error is given by

$$\varepsilon = R(0) - \sum_{k=1}^{p} \tilde{a}_k R(k) = 51 - 0.963 * 45 + 0.039 * 39 = 9.186.$$

3. a) A speech processing device for use in aircraft is being designed to include a noise reduction algorithm. The clean speech is denoted $s(n)$, the noisy speech is $y(n)$, and the noise is $\upsilon(n)$.

i) Give a brief summary of the operating principles of the spectral subtraction approach for noise reduction. Include a labelled block diagram of the spectral subtraction approach and a few sentences of explanation. *(Mathematical formulae or other analytical details are not required).* [3]

Solution:
Answers should include a diagram of this nature. The explanation should introduce the concept of a noise estimation step followed by a subtraction step.



ii) Describe and explain the minimum statistics approach for noise estimation. Include relevant mathematical formulae in your description. Begin by defining the input and output signals of the minimum statistics approach. [5]

Solution:
This technique is based on the assumption that, at some frequency, during a speech pause, or within brief periods between words and even syllables, the speech energy is close to zero. As a result, a short-term power spectrum estimate of the noisy signal, even during speech activity, decays frequently due to the noise power. Thus, by tracking the temporal spectral minimum without distinguishing between speech presence and speech absence, the noise power in a specific frequency band can be estimated.

The noisy speech is given by $y(n) = s(n) + \nu(n)$. Then the variance is obtain as

$$\hat{\phi}_y(l,k) = \alpha\hat{\phi}_y(l-1,k) + (1-\alpha)|Y(l,k)|^2.$$

The noise can then be estimated as

$$\hat{\phi}_\nu(l,k) = \min\left\{\hat{\phi}_y(l,k), \hat{\phi}_y(l-1,k), \ldots, \hat{\phi}_y(l-D+1,k)\right\}.$$

iii) Obtain an expression for the filter $H(l,k)$, for time frame l and frequency index k, that implements amplitude spectral subtraction exploiting a noise estimate. [4]

Solution:
$A_z(l,k) = A_y(l,k) - \sqrt{\hat{\phi}_v(k,k)}$

$Z(l,k) = H(l,k)Y(l,k)$ with $H(l,k) = 1 - \sqrt{\frac{\hat{\phi}_v(l,k)}{|Y(l,k)|^2}}$.

b) It is desired to perform a time-frequency analysis of a speech signal $s(n)$, with time index n, to give a spectrogram S.

i) Give the formula for the spectrogram S and define all variables and indices.

ii) Write 2 or 3 sentences to explain the processing employed in spectrographic analysis.

iii) Explain how the frequency resolution of spectrographic analysis is determined.

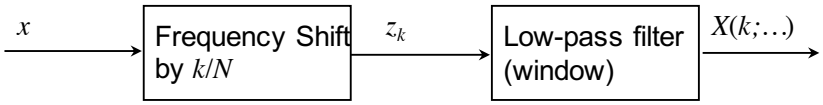
iv) Choosing a particular single frequency, explain the effect of any windowing applied. Support your explanation with any appropriate formulae and/or diagrams. [8]

Solution:
(i)

$$S(k,m) = \sum_{i=0}^{N-1} w(i)s(m-i)\exp\left(-j\frac{2\pi}{N}k(m-i)\right)$$

where m and k are time and frequency indices, respectively, and i is a dimensionless counter.

(ii)
$S(k,m)$ is a time-frequency analysis centred at sample $m-(N-1)/2$. The $(m-i)$ term in the exponent means that the phase origin remains consistent by cancelling out the linear phase shift introduced by a delay of m samples. m is the last sample in the analysis window, k is the frequency index, w is the window function.
(iii)
The frequency resolution is $1/N$ Hz (normalized units). There are $N/2+1$ independent frequency values.
(iv)
At a particular k, the kth frequency bin is a filtered version of $z_k(r) = s(r)\exp(-j\frac{2\pi}{N}kr)$ in which the filter has an impulse response of $w(i)$. The term $z_k(r)$ is a frequency shifted version of $s(r)$.



4. a) When considering statistical methods of automatic speech recognition, explain why it is usually preferred to employ log probability densities instead of probability densities. [3]

Solution:
Usually, many numbers less than one will be multiplied when many probabilities are multiplied together. The result of such a computation will often be an extremely small number that may underflow the computer's numerical representation capability. The log enables a much larger numerical dynamic range to be used successfully.

b) A speech utterance is processed by a statistical speech recognition system using hidden Markov models. The speech signal is divided into $T = 5$ frames. The duration of each frame t is 20 ms. The set of frames is therefore written $\mathbf{x}_t$ for $t = 1, 2, \ldots, T$.

The hidden Markov model has $S = 4$ states, numbered 1, 2, 3 and 4. The output log probability densities for frame t in state i are $\log(d_i(\mathbf{x}_t))$ and are given in Table 1. The transition probability from state i to state j of the model is a_{ij} as shown in Fig. 4.1.

	Input Frame				
State	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_4$	$\mathbf{x}_5$
1	-4	-8	-5	-5	-7
2	-2	-5	-7	-7	-6
3	-5	-3	-3	-8	-6
4	-6	-7	-8	-4	-8

Table 1: Output log probability densities

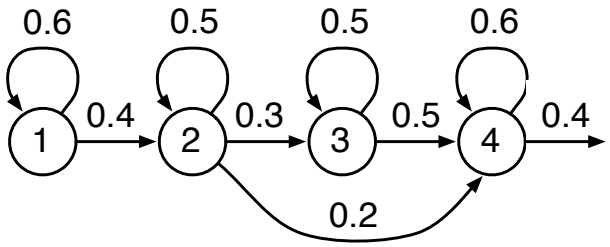


Figure 4.1: Hidden Markov model

i) For the case where

$$d_i(\mathbf{x}) = (2\pi)^{-1/2P} |\mathbf{C}_i|^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x}-\mathbf{m}_i)^T \mathbf{C}_i^{-1} (\mathbf{x}-\mathbf{m}_i)\right)$$

write a simplified expression for $\log(d_i(\mathbf{x}))$. Explain any advantageous features in the computation involved in calculating $\log(d_i(\mathbf{x}))$. [4]

Solution:

$$\log(d_i(\mathbf{x})) = -\frac{1}{2}\left(P\log(2\pi) + \log(|\mathbf{C}_i|) + (\mathbf{x}-\mathbf{m}_i)^T\mathbf{C}_i^{-1}(\mathbf{x}-\mathbf{m}_i)\right)$$

The first two terms are independent of $\mathbf{x}$ and only need to be calculated once. The terms involving the log are additive.

ii) The maximum probability density that the model generates frames $\mathbf{x}_1, \ldots, \mathbf{x}_t$ from any sequence of states for which frame 1 is in state 1 and frame t is in state s is defined as $B(t,s)$.

For the speech recognition task described above,

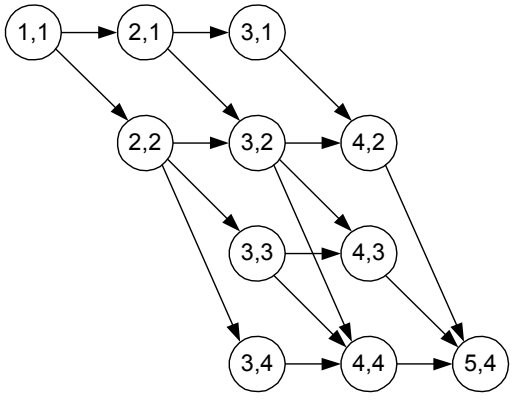
- write out a table showing the transition log probabilities from every state to every other state;
- draw the alignment lattice beginning with frame 1 in state 1, and ending with frame 5 in state 4;
- hence determine $\log(B(5,4))$ and the most likely state sequence.

[9]

Solution:
The table of log transition probabilities:

	1	2	3	4	end
1	-0.511	-0.916			
2		-0.693	-1.204	-1.609	
3			-0.693	-0.693	
4				-0.511	-0.916

The alignment lattice can be constructed as:



The table of log probabilities is then given by

B11	logd(1,1)	**-4**
B21	B11 + loga(1,1) + logd(1,2) = -12.5108	-12.5108
B22	B11 + loga(1,2) + logd(2,2) = -9.9163	**-9.9163**
B31	B21 + loga(1,1) + logd(1,3) = -18.0217	-18.0217
B32	B21 + loga(1,2) + logd(2,3)= -20.4271	-17.6094
	B22 + loga(2,2) + logd(2,3) = **-17.6094**	
B33	B22 + loga(2,3) + logd(3,3) = -14.1203	**-14.1203**
B34	B22 + loga(2,4) + logd(4,3) = -19.5257	-19.5257
B42	B31 + loga(1,2) + logd(2,4) = -25.9379	-25.3026
	B32 + loga(2,2) + logd(2,4) = **-25.3026**	
B43	B32 + loga(2,3) + logd(3,4) = -26.8134	-22.8134
	B33 + loga(3,3) + logd(3,4) = **-22.8134**	
B44	B32 + loga(2,4) + logd(4,4) = -23.2189	**-18.8134**
	B33 + loga(3,4) + logd(4,4) = **-18.8134**	
	B34 + loga(4,4) + logd(4,4) = -24.0366	
B54	B42 + loga(2,4) + logd(4,5) = -34.9120	**-27.3242**
	B43 + loga(3,4) + logd(4,5) = -31.5066	
	B44 + loga(4,4) + logd(4,5) = **-27.3242**	

The corresponding path is therefore 1,2,3,4,4.

c) Consider an alternative speech recognition system based on Deep Neural Networks (DNNs). Explain a common training strategy to improve the robustness of trained DNN models for speech recognition in real-world environments. [4]

Solution:
Data augmentation is a common strategy to increase the quantity of training data, and improve the robustness of the trained DNN models, exploiting for example:

- changing the level of the signal
- changing the speaking rate
- adding noise
- adding reverb by convolving with a range of room impulse responses
- passing the speech signal through a speech CODEC, or various CODECs, a different data rates
- generating new speech using text-to-speech

The performance improvement comes from improving the chances that the trained model matches well with the test data at inference.